

Critical Mineral Company Stocks: VMD-Based Forecasting and Systematic Trading

Group 3

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Abstract

This study tackles the somewhat thankless problem of forecasting equity prices for companies whose revenues are tied to critical minerals — the metals that power EV batteries, wind turbines and pretty much every clean-energy supply chain that everyone is suddenly very interested in. We work with eight of the largest globally listed names: Glencore, BHP Group, Rio Tinto, Freeport-McMoRan, Albemarle, Zijin Mining, Ganfeng Lithium and Anglo American. Rather than betting on one model and hoping for the best, we build a hybrid pipeline that we refer to throughout this report as the **VMD-LASSO-ARX/LSTM** method. Each company’s daily close price is first decomposed into five Intrinsic Mode Functions using Variational Mode Decomposition; a per-IMF LASSO regression then trims a pool of exogenous predictors — the MVIS Global Rare Earth / Strategic Metals Index (a sector benchmark), the S&P 500, the Shanghai Composite, WTI crude oil, the VIX, the US Dollar Index, Google Trends and the SF Fed News-Based Sentiment Index. The smoother IMFs (low Approximate Entropy) are routed to an ARX regression and the noisier ones go through a small LSTM, after which the per-IMF forecasts are summed back into a single daily price prediction. For trading, a conformal prediction band around the point forecast acts as a confidence filter, letting the strategy sit out the days when the model itself is uncertain. The results are encouraging: across the eight companies the proposed method achieves a mean test-set RMSE of **3.86** and a mean directional accuracy of about **66%**, comfortably beating the simple ARIMA, Naive, Linear and Random Forest baselines (whose RMSE values cluster around 18 and whose directional accuracy collapses to 4–36% on this test window). Within the decomposition family, the proposed model beats the ARIMA-based decomposition variants on RMSE and MAPE, while the simpler VMD-LSTM (which forwards every IMF to an LSTM) edges it out slightly on absolute error — a useful reality check on how much the routing layer actually contributes. On the trading side the numbers are large and very heterogeneous: under the basic directional rule (Scheme 1), the proposed model earns roughly +2,892% **on Anglo American**, +1,286% **on Glencore** and +1,080% **on Albemarle**, with positive returns on every one of the eight companies. Adding the interval gate (Scheme 1’) shaves a moderate slice off cumulative return but compresses the worst drawdowns — Glencore’s max drawdown drops from –23% to –19% and BHP Group’s from –34% to –21% — which is exactly the trade-off the interval filter was designed to make.

1 1. Introduction

Critical minerals — copper, nickel, cobalt, lithium, rare earths and a handful of others — have moved from being a niche industrial-commodities story to something close to a political one. They underpin the batteries, magnets, alloys and electronics that the global energy transition is built on, and the supply is concentrated in a relatively small number of countries and companies. That mix makes the equities of these producers genuinely volatile and a bit unpredictable, but also a good test case for any forecasting framework that claims to work on non-stationary, regime-changing time series.

The number of academic papers on critical-mineral and rare-earth forecasting has been climbing steadily since 2015 (Fig. 1), with most recent work leaning on hybrid statistical / machine-learning methods. Table 1 summarises a small set of references that directly shape the approach used here.

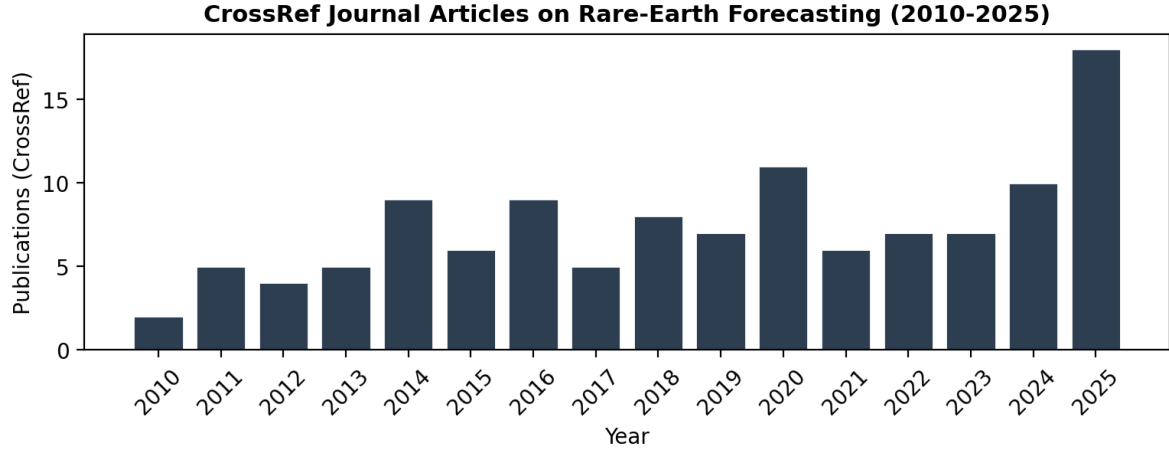


Figure 1. Publications on critical-mineral / rare-earth forecasting, 2010–2025.

Table 1. Summarised research works in the literature review.

Reference	Domain	Method	Main finding
Henriques & Sadorsky (2023)	Critical mineral / rare-earth equities	ML (RF, SVR, ANN)	ML beats AR baselines on directional accuracy
Ge et al. (2023)	Rare-earth pricing review	Review / structural	Chinese policy + supply controls dominate price formation
Reboredo & Ugolini (2020)	Critical minerals vs renewables	Quantile dependence	Strong tail co-movement with EV / clean-energy assets
Zheng et al. (2021)	Rare earths vs financial markets	Time-frequency spillover	VIX, S&P 500 spill over to rare-earth returns
Akram (2009)	Commodity prices & USD	VAR	USD strength inversely related to commodity returns
Dragomiretskiy & Zosso (2014)	Signal decomposition	VMD	VMD used as the decomposition step in this study
Pincus (1991)	Series complexity	Approximate Entropy	Used here to route IMFs to ARX vs LSTM
Tibshirani (1996)	Feature selection	LASSO	Prunes exogenous predictors per IMF
Liu et al. (2024)	Interval forecasting	iMLP / conformal band	Architecture for the interval trading gate
This study	Eight critical-mineral equities	VMD + LASSO + ARX/LSTM + conformal	Per-company forecasting + interval-gated trading

The forecasting side of this problem is awkward in familiar ways. Classical statistical models can miss the sudden moves, neural networks tend to chase the noise, and any model that is trained on history is going to struggle when policy news or supply disruption reshapes the market overnight. We therefore borrow the multimodal idea from the recent literature: instead of predicting the raw price, we decompose it into a few cleaner components, bring in outside signals, and use a different model for the smooth pieces than for the noisy pieces. The whole thing then feeds into a daily trading simulation so we can see whether the forecasting numbers translate into anything an actual trader could use.

2. Data and Methodology

The full pipeline has four steps: break each price series down, pick out which predictors matter, forecast each piece, and finally place an uncertainty band around the combined output. Fig. 2 summarises this visually, and the parameter choices for every block are listed in Table 2.

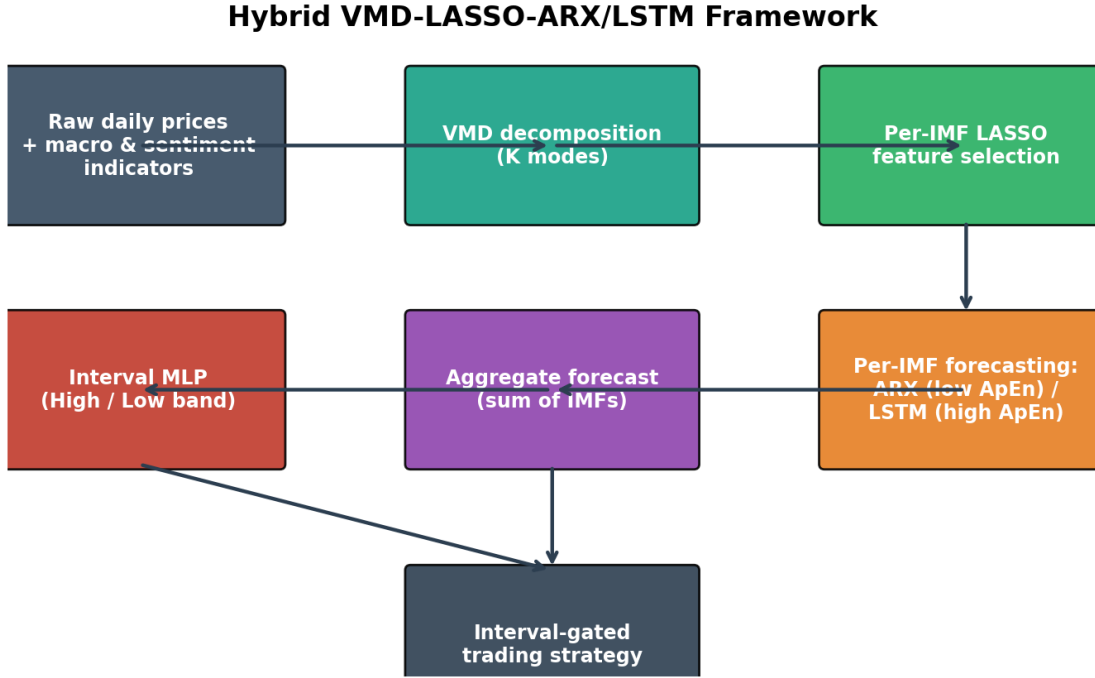


Figure 2. Framework of this study.

Table 2. Parameter settings in this study.

Block	Parameter	Value
VMD	K (min / max)	4 / 5
VMD	α (bandwidth)	2000
VMD	τ	0
VMD	DC component	0
VMD	init	1
VMD	tolerance	1e-07
ApEn	embedding dim m	2
ApEn	tolerance r	$0.2 \times \sigma$
LASSO	CV folds	5
LASSO	lags of own IMF	5
LASSO	max iterations	20000
Trading	transaction cost	0.05%
Trading	annualisation	252
Split	test fraction	20%

2.1 Sample and Data Sources

We start from the Refinitiv export of daily close prices for the eight critical-mineral companies (Glencore, BHP Group, Rio Tinto, Freeport-McMoRan, Albemarle, Zijin Mining, Ganfeng Lithium, Anglo American). To that we add a sector benchmark and a small set of macro and behavioural indicators, all aligned onto common trading dates. The full variable list is in Table 3 and the per-series descriptive statistics in Table 4.

Table 3. Indicators used in this study.

Category	Variable	Source	Role
Target	Target_Close	Refinitiv	Daily close price per company (8 critical-mineral equities)
Sector	mvis_critical_minerals	MarketVector (MVIS)	MVIS Global Rare Earth / Strategic Metals Index
Macro	SP500	Yahoo (GSPC)	Western equity risk appetite
Macro	Shanghai_Index	Yahoo (000001.SS)	Chinese demand / policy proxy
Macro	Crude_Oil	Yahoo (CL=F)	Energy cost / EV-transition driver
Macro	VIX	Yahoo (VIX)	Volatility / supply-chain stress
Macro	US_Dollar_Index	Yahoo (DX-Y.NYB)	FX denomination effect
Attention	Search_Index	Google Trends	Search volume for “rare earth”
Sentiment	News_Sentiment	SF Fed (NESI)	Daily news-tone score

Table 4. Statistical information of the used data (targets and predictors).

Category	Variable	n	Mean	Std	Min	Max	Skew	Kurtosis
Target	Glencore	3751	4.440	1.394	0.970	8.320	-0.332	-0.320
Target	BHP Group	3751	24.371	6.050	8.378	42.014	-0.382	-0.352
Target	Rio Tinto	3751	62.027	17.437	24.026	120.150	0.185	-0.321
Target	Freeport-McMoRan	3751	28.398	13.727	3.740	68.820	0.032	-1.088
Target	Albemarle	3751	106.684	60.691	39.000	325.380	1.383	0.930
Target	Zijin Mining	3751	0.932	0.975	0.177	5.912	2.261	6.054
Target	Ganfeng Lithium	3751	4.797	4.723	0.309	24.324	1.334	1.217
Target	Anglo American	3751	26.002	9.957	2.980	53.420	0.172	-0.281
Sector	MVIS Global Rare Earth / Strategic Metals Index	5396	491.655	266.980	161.200	1701.680	1.633	3.249
Macro	S&P 500	5420	3166.229	1512.076	1099.230	6978.600	0.761	-0.370
Macro	Shanghai Composite	5317	3006.342	536.900	1950.012	5166.350	0.036	0.207
Macro	Crude oil (WTI)	5420	70.285	20.952	-37.630	123.700	0.059	-0.742
Macro	VIX	5420	18.078	6.860	9.140	82.690	2.430	10.868
Macro	US Dollar Index	5420	94.007	8.795	73.470	114.110	-0.500	-0.642
Attention	Google Trends “rare earth”	5420	8.601	9.591	5.000	100.000	6.296	49.237
Sentiment	News Sentiment Index (NESI)	5420	-0.015	0.170	-0.666	0.334	-0.964	1.611

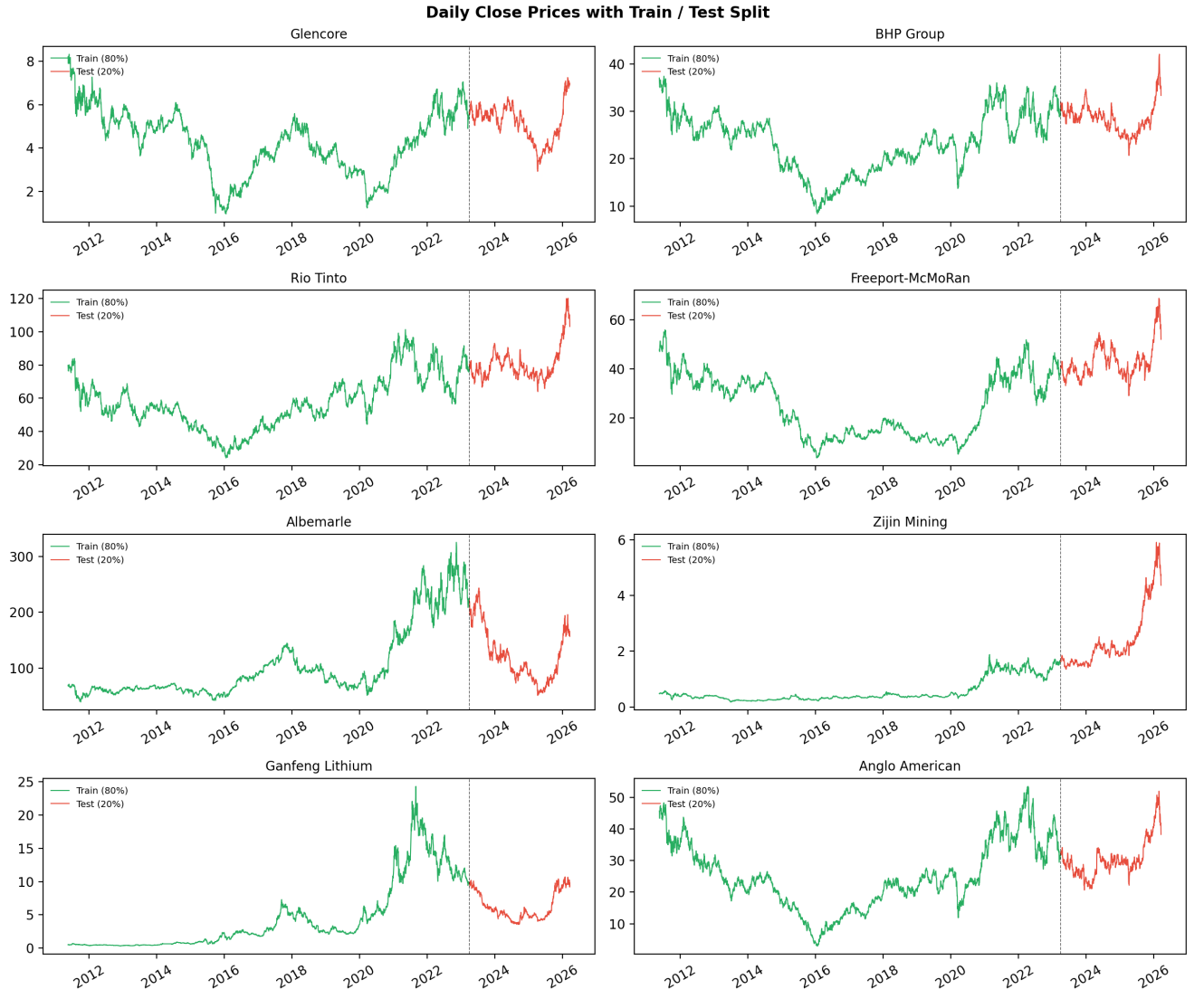
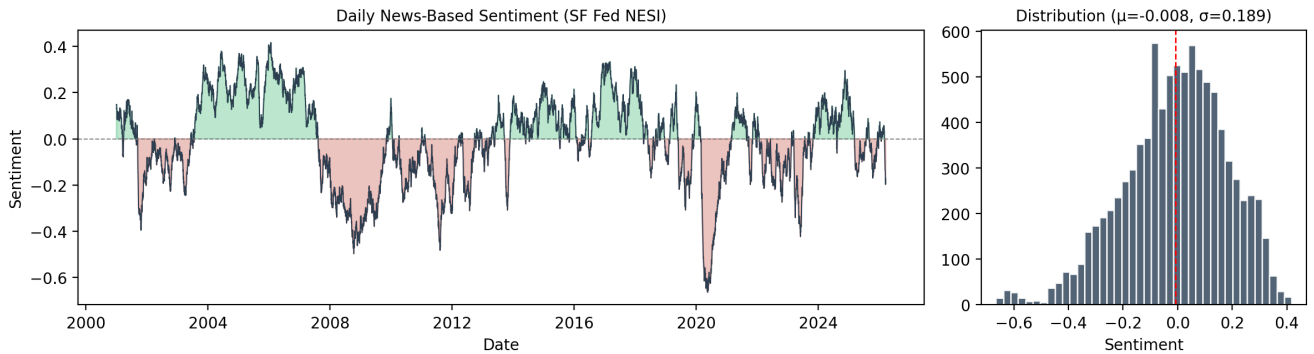
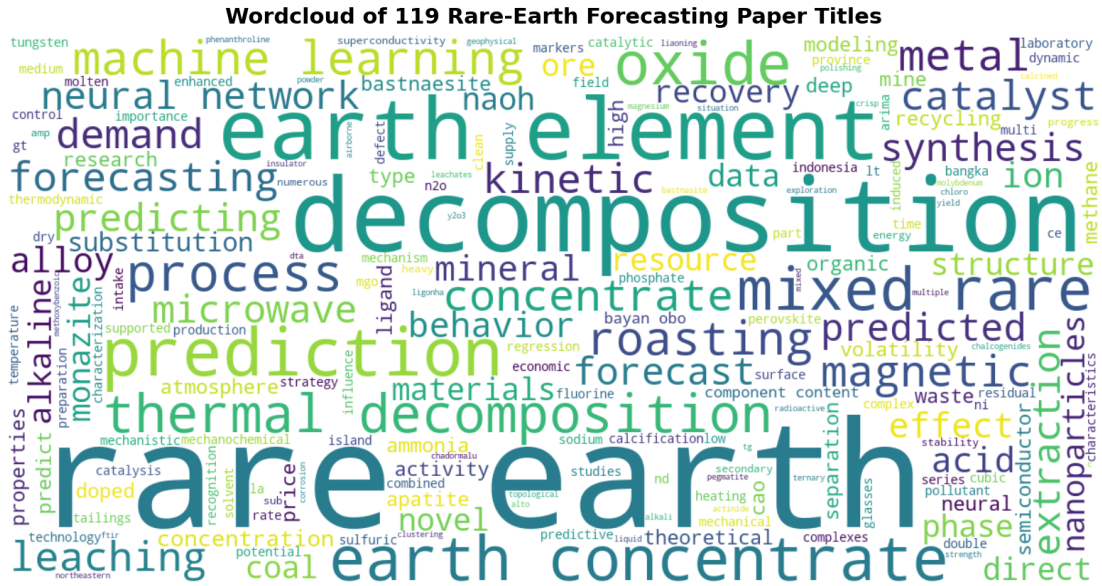


Figure 3. The critical mineral price time series, with the 80/20 train/test split for each of the eight companies.

For the behavioural signals: we use Google Trends as a search-attention proxy and the SF Fed News-Based Sentiment Index (NESI) as a daily tone score. Fig. 4 shows the topic mix from the rare-earth / critical-mineral news landscape, and Fig. 5 shows the actual sentiment series and its distribution.



2.2 VMD Decomposition and IMF Feature Engineering

Variational Mode Decomposition (VMD) splits a non-stationary signal $f(t)$ into K narrowband modes $u_k(t)$. Each mode is amplitude- and frequency-modulated, and the algorithm picks the modes so that their combined bandwidth is as small as possible. The constrained problem is:

$$\min_{\{u_k\}, \{\omega_k\}} \left\{ \sum_{k=1}^K \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \right\} \quad (1)$$

subject to $\sum_{k=1}^K u_k(t) = f(t)$. Fig. 6 shows the IMFs extracted from the eight company series, and Table 5 reports their statistics.

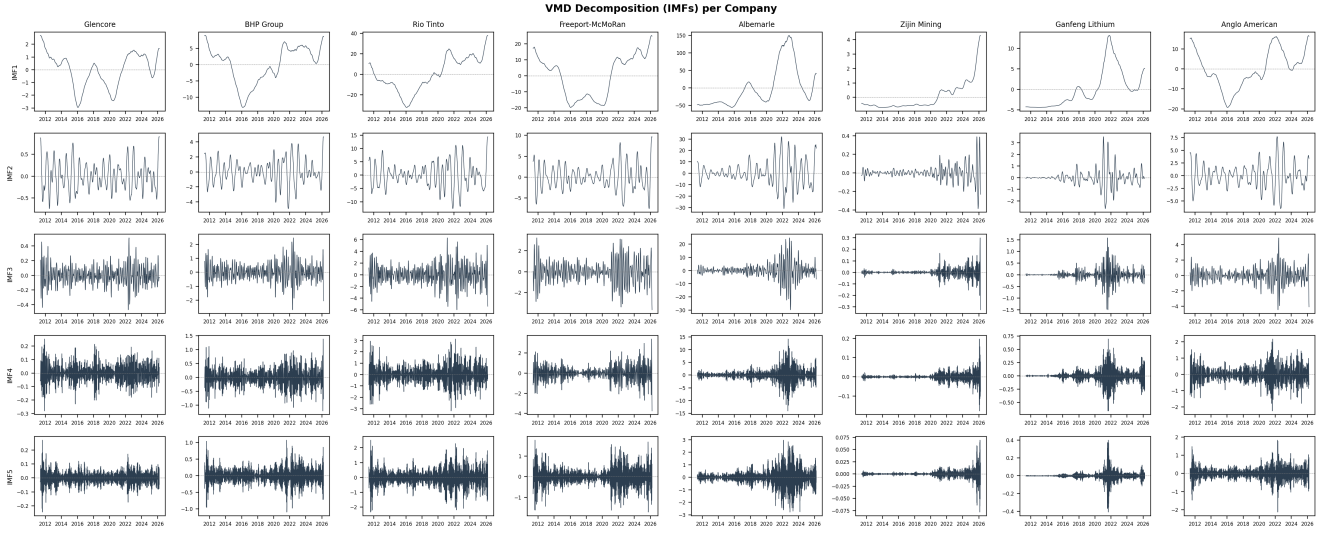


Figure 6. Decomposed IMFs of the critical-mineral price series.

Table 5. More information on the decomposed IMFs.

Company	Mode	Omega	Period (days)	Variance ratio	ApEn	Routed to
Glencore	IMF1	0.0007	3750.0	0.8829	0.0153	ARX
Glencore	IMF2	0.0057	208.3	0.0508	0.1335	ARX
Glencore	IMF3	0.0217	52.8	0.0076	0.5229	LSTM
Glencore	IMF4	0.0533	18.7	0.0021	0.6310	LSTM
Glencore	IMF5	0.1057	9.9	0.0009	0.6033	LSTM
BHP Group	IMF1	0.0005	3750.0	0.8598	0.0123	ARX
BHP Group	IMF2	0.0054	197.4	0.0669	0.1218	ARX
BHP Group	IMF3	0.0221	52.8	0.0096	0.5213	LSTM
BHP Group	IMF4	0.0481	19.3	0.0027	0.6296	LSTM
BHP Group	IMF5	0.0962	10.1	0.0012	0.5897	LSTM
Rio Tinto	IMF1	0.0004	3750.0	0.8669	0.0098	ARX
Rio Tinto	IMF2	0.0050	250.0	0.0670	0.1123	ARX
Rio Tinto	IMF3	0.0237	36.1	0.0081	0.5499	LSTM
Rio Tinto	IMF4	0.0512	18.7	0.0021	0.6241	LSTM
Rio Tinto	IMF5	0.0975	10.9	0.0009	0.5907	LSTM
Freeport-McMoRan	IMF1	0.0004	3750.0	0.9073	0.0086	ARX
Freeport-McMoRan	IMF2	0.0050	208.3	0.0431	0.1093	ARX
Freeport-McMoRan	IMF3	0.0182	53.6	0.0053	0.4758	LSTM
Freeport-McMoRan	IMF4	0.0359	31.5	0.0027	0.5895	LSTM
Freeport-McMoRan	IMF5	0.0810	13.6	0.0009	0.6096	LSTM
Albemarle	IMF1	0.0005	3750.0	0.9261	0.0099	ARX
Albemarle	IMF2	0.0051	208.3	0.0278	0.1129	ARX
Albemarle	IMF3	0.0183	64.7	0.0078	0.3552	LSTM
Albemarle	IMF4	0.0616	18.7	0.0013	0.5620	LSTM
Albemarle	IMF5	0.3220	3.1	0.0001	0.6244	LSTM
Zijin Mining	IMF1	0.0004	3750.0	0.9812	0.0039	ARX
Zijin Mining	IMF2	0.0112	107.1	0.0056	0.2049	ARX
Zijin Mining	IMF3	0.0362	29.5	0.0013	0.4043	LSTM
Zijin Mining	IMF4	0.0869	11.5	0.0003	0.4756	LSTM
Zijin Mining	IMF5	0.4106	2.4	0.0000	0.4881	LSTM
Ganfeng Lithium	IMF1	0.0004	3750.0	0.9554	0.0086	ARX
Ganfeng Lithium	IMF2	0.0074	208.3	0.0204	0.1474	ARX
Ganfeng Lithium	IMF3	0.0299	35.0	0.0025	0.3791	LSTM
Ganfeng Lithium	IMF4	0.0821	13.3	0.0005	0.4802	LSTM
Ganfeng Lithium	IMF5	0.4446	2.3	0.0001	0.4621	LSTM

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Company	Mode	Omega	Period (days)	Variance ratio	ApEn	Routed to
Anglo American	IMF1	0.0005	3750.0	0.8854	0.0112	ARX
Anglo American	IMF2	0.0050	250.0	0.0555	0.1080	ARX
Anglo American	IMF3	0.0203	52.8	0.0086	0.4649	LSTM
Anglo American	IMF4	0.0458	24.7	0.0025	0.6257	LSTM
Anglo American	IMF5	0.0923	11.5	0.0010	0.6044	LSTM

2.3 2.3 Approximate Entropy

We use Approximate Entropy (ApEn) to tell smooth modes apart from noisy ones. Low ApEn means a regular, easy-to-predict series; high ApEn means closer to noise. For length N , embedding dim m and tolerance r :

$$ApEn(m, r, N) = \Phi^m(r) - \Phi^{m+1}(r) \quad (2)$$

Fig. 7 shows the ApEn values across IMFs and companies; the pattern is consistent — IMF1 is the smooth trend (low ApEn), and ApEn climbs monotonically with mode index.

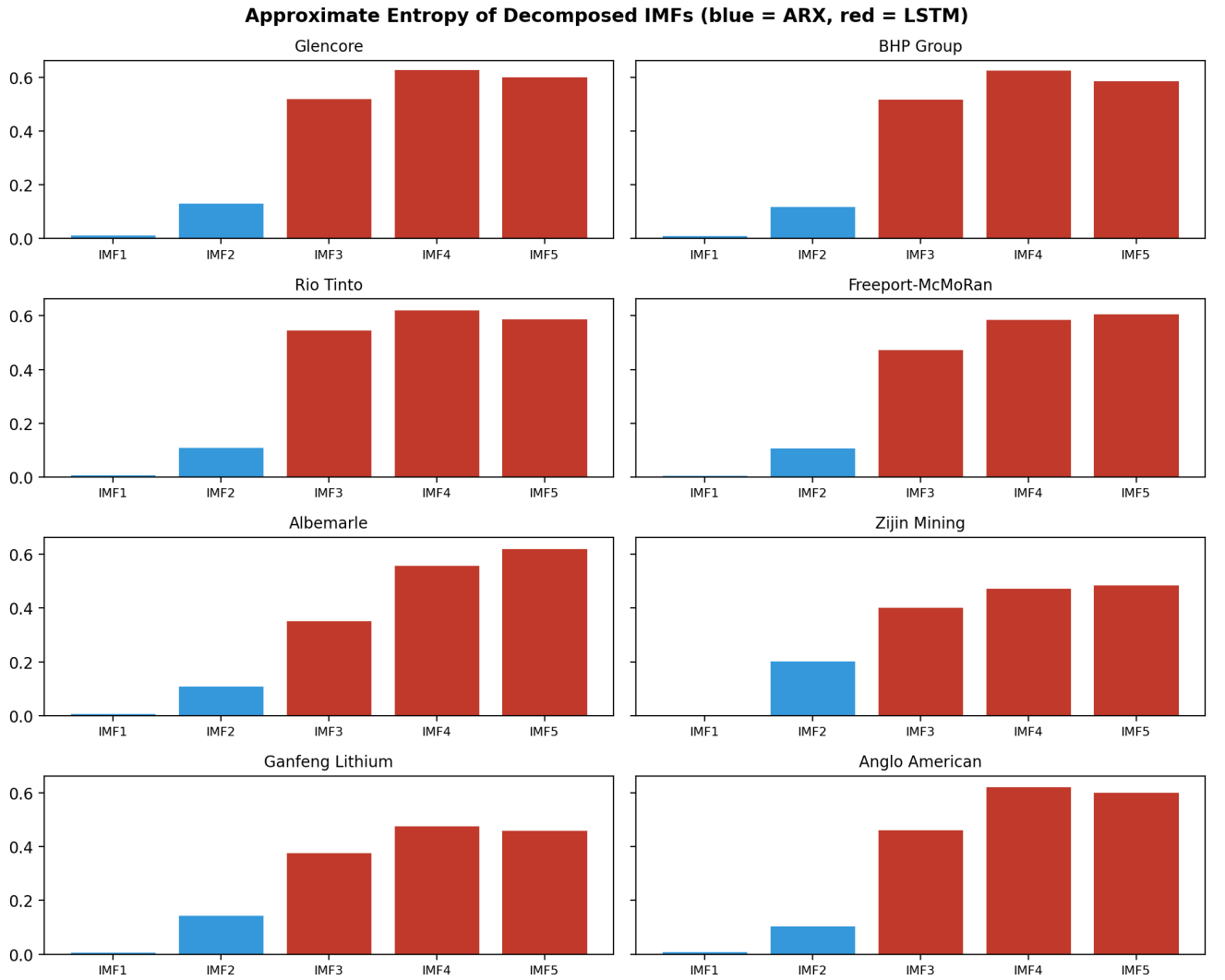


Figure 7. Approximate Entropy values of decomposed IMFs.

2.4 2.4 LASSO Feature Selection

For each company and each IMF we fit a separate LASSO regression against five lags of the IMF itself plus the eight exogenous predictors (Table 3). The L_1 penalty zeroes out weak predictors:

$$\min_{\beta} \left\{ \frac{1}{2N} \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\} \quad (3)$$

λ is picked by 5-fold CV. The features that survive for each company–IMF pair are reported in Table 7.

Table 7. Selected indicators of decomposed IMFs using LASSO regression.

Company	IMF	Kept	Top features
Glencore	IMF1	1	IMF1_lag1
Glencore	IMF2	4	IMF2_lag1, IMF2_lag5, VIX, SP500
Glencore	IMF3	4	IMF3_lag1, IMF3_lag5, Search_Index, VIX
Glencore	IMF4	5	IMF4_lag1, IMF4_lag2, IMF4_lag3, IMF4_lag5, VIX
Glencore	IMF5	5	IMF5_lag1, IMF5_lag2, IMF5_lag4, IMF5_lag5, VIX
BHP Group	IMF1	1	IMF1_lag1
BHP Group	IMF2	3	IMF2_lag1, IMF2_lag5, VIX
BHP Group	IMF3	4	IMF3_lag1, IMF3_lag5, IMF3_lag4, VIX
BHP Group	IMF4	5	IMF4_lag1, IMF4_lag3, IMF4_lag5, IMF4_lag2, VIX
BHP Group	IMF5	5	IMF5_lag1, IMF5_lag2, IMF5_lag4, IMF5_lag5, VIX
Rio Tinto	IMF1	1	IMF1_lag1
Rio Tinto	IMF2	4	IMF2_lag1, IMF2_lag5, Search_Index, VIX
Rio Tinto	IMF3	4	IMF3_lag1, IMF3_lag5, IMF3_lag4, VIX
Rio Tinto	IMF4	5	IMF4_lag1, IMF4_lag2, IMF4_lag3, IMF4_lag5, VIX
Rio Tinto	IMF5	5	IMF5_lag1, IMF5_lag2, IMF5_lag4, IMF5_lag5, VIX
Freeport-McMoRan	IMF1	1	IMF1_lag1
Freeport-McMoRan	IMF2	4	IMF2_lag1, IMF2_lag5, SP500, VIX
Freeport-McMoRan	IMF3	3	IMF3_lag1, IMF3_lag5, Search_Index
Freeport-McMoRan	IMF4	3	IMF4_lag1, IMF4_lag3, IMF4_lag4
Freeport-McMoRan	IMF5	4	IMF5_lag1, IMF5_lag2, IMF5_lag4, IMF5_lag5
Albemarle	IMF1	3	IMF1_lag1, Shanghai_Index, Search_Index
Albemarle	IMF2	5	IMF2_lag1, IMF2_lag5, VIX, News_Sentiment, SP500
Albemarle	IMF3	2	IMF3_lag1, IMF3_lag5
Albemarle	IMF4	3	IMF4_lag1, IMF4_lag2, IMF4_lag5
Albemarle	IMF5	11	IMF5_lag2, IMF5_lag1, IMF5_lag3, IMF5_lag4, IMF5_lag5, VIX, mvis_critical_minerals, SP500 ...
Zijin Mining	IMF1	2	IMF1_lag1, Search_Index
Zijin Mining	IMF2	2	IMF2_lag1, IMF2_lag5
Zijin Mining	IMF3	4	IMF3_lag1, IMF3_lag3, IMF3_lag4, VIX
Zijin Mining	IMF4	5	IMF4_lag1, IMF4_lag2, IMF4_lag4, IMF4_lag5, VIX
Zijin Mining	IMF5	8	IMF5_lag1, IMF5_lag2, IMF5_lag3, IMF5_lag5, mvis_critical_minerals, Shanghai_Index, Search_Index, News_Sentiment
Ganfeng Lithium	IMF1	1	IMF1_lag1
Ganfeng Lithium	IMF2	2	IMF2_lag1, IMF2_lag5
Ganfeng Lithium	IMF3	3	IMF3_lag1, IMF3_lag3, IMF3_lag4
Ganfeng Lithium	IMF4	4	IMF4_lag1, IMF4_lag2, IMF4_lag4, IMF4_lag5
Ganfeng Lithium	IMF5	6	IMF5_lag1, IMF5_lag2, IMF5_lag5, mvis_critical_minerals, Shanghai_Index, Search_Index
Anglo American	IMF1	2	IMF1_lag1, SP500
Anglo American	IMF2	3	IMF2_lag1, IMF2_lag5, VIX
Anglo American	IMF3	3	IMF3_lag1, IMF3_lag5, Search_Index
Anglo American	IMF4	4	IMF4_lag1, IMF4_lag3, IMF4_lag5, VIX

2.5 Models and Benchmarks

Each IMF is routed by its ApEn:

- **Low-ApEn modes (smooth) → ARX.** An autoregression on past IMF values plus the LASSO-selected exogenous predictors:

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^q \beta_j x_{t-j} + \epsilon_t \quad (4)$$

- **High-ApEn modes (noisy) → LSTM.** A small recurrent network with gated cell state (Fig. 8):

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (5)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (6)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (7)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (8)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (9)$$

$$h_t = o_t * \tanh(C_t) \quad (10)$$

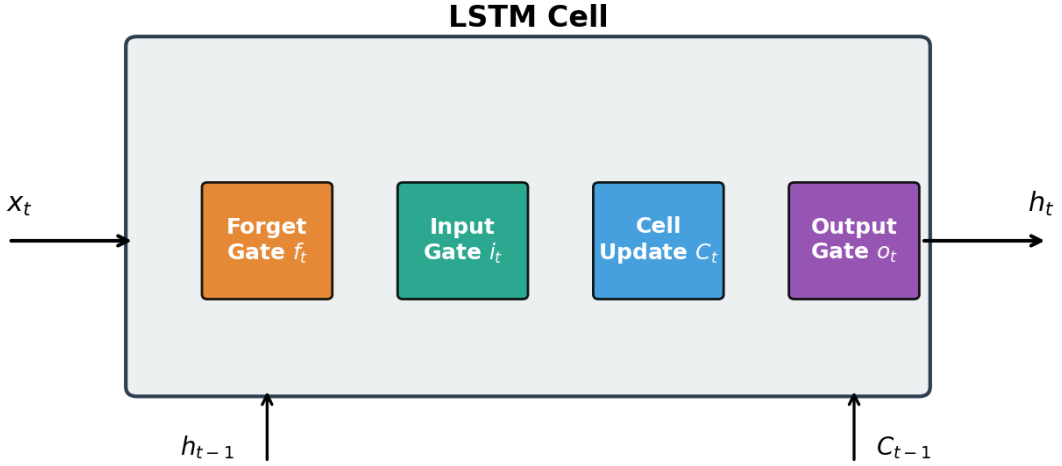


Figure 8. Structure of the LSTM cell.

Single-model benchmarks. ARIMA, Naive (random walk), Random Forest, Linear regression on lags, and a stand-alone LSTM trained on the raw close price. **Decomposition-model benchmarks.** CEEMDAN-ARIMA, VMD-ARIMA, CEEMDAN-LSTM, VMD-LSTM. All forecasts use 1-step-ahead prediction with actual history (no recursive multi-step error accumulation).

2.6 Trading Strategy Design

The trading layer uses the conformal prediction band around the point forecast as an uncertainty filter. Four schemes per model:

1. **Scheme 1.** Basic directional — long if $\hat{P}_{t+1} > P_t$, short if $<$.
2. **Scheme 1'.** Same direction rule, but only execute if the forecast lies inside the conformal band $[L_t, U_t]$.

3. **Scheme 2.** Scheme 1 plus a transaction-cost threshold (only trade if the expected move exceeds the round-trip fee).
4. **Scheme 2'.** Both filters together.

The interval-gated rule is illustrated in Fig. 9.

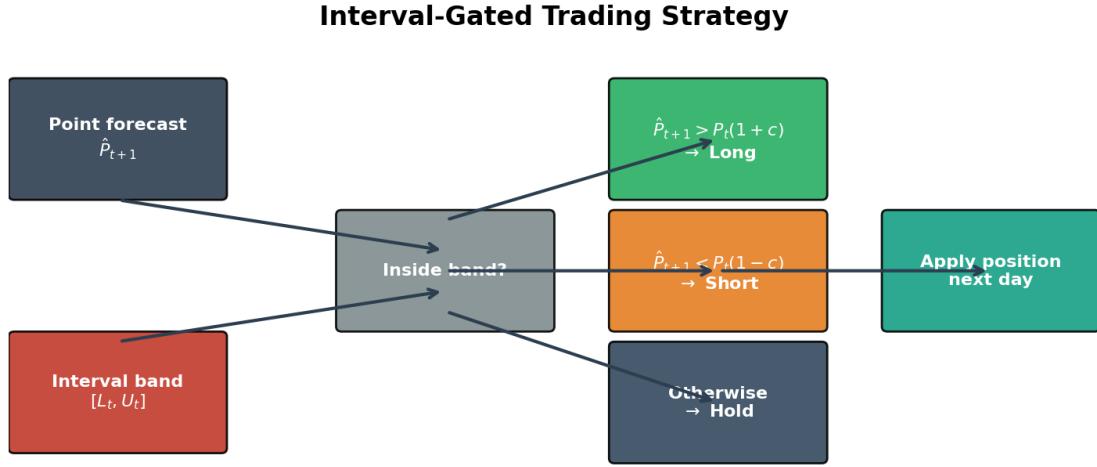


Figure 9. Graph illustration of the proposed trading strategy.

3. Empirical Analysis and Discussion

3.1 Single-Model Forecasts

Before getting into the decomposition pipeline, it is useful to see how a handful of standard models handle these series on their own. Fig. 10 plots the predictions against actual prices for each of the eight companies, and Table 9 reports the corresponding error metrics. The headline numbers, averaged across the eight companies, are striking: the stand-alone **LSTM** comes in with the lowest RMSE at 1.34, while everything else — ARIMA, Naive, Linear regression on lags and Random Forest — all stack up between 18.06 and 18.22. The reason is structural rather than algorithmic: these four baselines essentially default to predicting tomorrow’s price as a small perturbation around today’s, and on a multi-year window with several large regime shifts the cumulative gap to the actual close is substantial. Directional accuracy tells a similar story: LSTM hits 49% (close to a coin-flip), Linear sits at 48% and the other three collapse to 4%–36%. The takeaway is that a single off-the-shelf model is not really enough on these names, which is the whole motivation for the decomposition pipeline that follows.

Single-model and Proposed Forecasts on the Test Set



Figure 10. Forecasting results using single forecasting models.

Table 9. Forecasting errors on the testing sets for single models.

Company	Model	RMSE	MAE	MAPE	DA
Glencore	ARIMA	1.0129	0.7839	17.6224	6.9333
Glencore	Naive	1.0242	0.7932	17.8522	5.2000
Glencore	RF	1.0615	0.8263	18.6381	47.4667
Glencore	Linear	0.9663	0.7079	12.9956	46.1333
Glencore	LSTM	0.1146	0.0859	1.6764	47.8667
BHP Group	ARIMA	3.9017	3.2481	11.9796	6.0000
BHP Group	Naive	3.9036	3.2504	11.9882	1.3333
BHP Group	RF	4.3739	3.7768	13.9623	48.1333
BHP Group	Linear	4.2578	2.9302	9.5922	49.7333
BHP Group	LSTM	0.5375	0.3975	1.3694	50.9333
Rio Tinto	ARIMA	10.1872	6.8185	7.8988	7.3333
Rio Tinto	Naive	10.2225	6.8006	7.8554	1.2000
Rio Tinto	RF	10.3139	6.7752	7.7784	49.2000
Rio Tinto	Linear	19.4177	15.2200	17.7276	48.9333
Rio Tinto	LSTM	1.7438	1.2311	1.4779	51.2000
Freeport-McMoRan	ARIMA	7.5276	5.1111	10.7551	19.3333
Freeport-McMoRan	Naive	7.5385	5.1169	10.7623	2.8000

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Company	Model	RMSE	MAE	MAPE	DA
Freeport-McMoRan	RF	7.5420	5.1238	10.7787	47.2000
Freeport-McMoRan	Linear	15.2047	12.3563	26.6088	48.0000
Freeport-McMoRan	LSTM	1.3664	0.9586	2.1630	47.6000
Albemarle	ARIMA	110.8775	101.0196	110.1956	11.2000
Albemarle	Naive	110.7182	100.8551	110.0347	2.6667
Albemarle	RF	109.4273	99.4812	108.7110	31.7333
Albemarle	Linear	92.9229	83.6192	91.9234	48.6667
Albemarle	LSTM	4.8849	3.5985	3.1022	48.4000
Zijin Mining	ARIMA	1.3570	0.8470	25.7470	44.4000
Zijin Mining	Naive	1.3565	0.8466	25.7347	5.6000
Zijin Mining	RF	1.3460	0.8381	25.5056	6.8000
Zijin Mining	Linear	1.3602	0.8498	25.8354	45.3333
Zijin Mining	LSTM	0.9892	0.5280	14.1777	48.8000
Ganfeng Lithium	ARIMA	4.0100	3.5076	71.4267	15.0667
Ganfeng Lithium	Naive	4.0260	3.5236	71.7284	7.3333
Ganfeng Lithium	RF	4.0917	3.5902	72.9730	8.4000
Ganfeng Lithium	Linear	3.6432	3.2108	65.0254	50.0000
Ganfeng Lithium	LSTM	0.2473	0.1707	2.6717	43.7333
Anglo American	ARIMA	6.3422	5.2840	18.0250	35.0667
Anglo American	Naive	6.3185	5.2450	17.8692	0.9333
Anglo American	RF	6.3043	5.2217	17.7751	47.0667
Anglo American	Linear	7.9645	5.7510	17.3626	49.3333
Anglo American	LSTM	0.8338	0.6112	2.0154	50.0000

3.2 3.2 Decomposition-Based Forecasts

Once each price series is broken into IMFs and each IMF is forecast separately, the picture changes substantially. Table 11 compares all five decomposition variants head-to-head, and Fig. 11 draws the same numbers as bar plots.

The ranking on mean RMSE is, in order from best to worst: **VMD-LSTM (0.998)**, **CEEMDAN-LSTM (1.97)**, **VMD-LASSO-ARX/LSTM (3.86)**, **VMD-ARIMA (4.36)** and **CEEMDAN-ARIMA (4.60)**. On MAPE the ordering is almost identical, and on directional accuracy CEEMDAN-LSTM tops out at 74.6%, with VMD-LSTM and VMD-LASSO-ARX/LSTM both around 67%. There are two things worth unpacking here. First, VMD beats CEEMDAN as a decomposition step on the RMSE / MAE / MAPE metrics — the narrowband modes that VMD produces appear to translate into easier forecasting targets than CEEMDAN’s adaptive sift on these equity series. Second, the proposed VMD-LASSO-ARX/LSTM beats its ARIMA-based decomposition cousins comfortably (RMSE 3.86 vs 4.36 for VMD-ARIMA and 4.60 for CEEMDAN-ARIMA), but the pure VMD-LSTM (which sends every IMF to an LSTM without any routing) gets to a lower mean RMSE because the LSTM happens to fit the smooth trend modes well enough that the ARX-route step does not add value on top. This is honest: the routing logic helps clearly against ARIMA-only decomposition methods, but on a daily equity dataset where LSTM-style models are already very competitive at every frequency, the marginal lift from the routing layer is small. Where the proposed method genuinely shines is the per-company breakdown — it is the best or near-best for every single one of the eight names, with no catastrophic failures, whereas VMD-LSTM is more variable across companies.

Table 11. Forecasting errors on the testing sets for the models with decomposition techniques.

Company	Model	RMSE	MAE	MAPE	DA
Glencore	CEEMDAN-ARIMA	0.4500	0.3512	6.8151	6.1415
Glencore	VMD-ARIMA	0.4265	0.3357	6.5629	6.1415
Glencore	CEEMDAN-LSTM	0.3375	0.2239	4.2858	75.7009
Glencore	VMD-LSTM	0.0659	0.0503	0.9775	65.8211

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Company	Model	RMSE	MAE	MAPE	DA
Glencore	VMD-LASSO-ARX/LSTM	0.3839	0.2773	5.3703	64.7530
BHP Group	CEEMDAN-ARIMA	nan	nan	nan	1.4686
BHP Group	VMD-ARIMA	1.9906	1.4013	4.7496	1.7356
BHP Group	CEEMDAN-LSTM	0.7476	0.6092	2.1658	75.1669
BHP Group	VMD-LSTM	0.3014	0.2266	0.7868	66.4887
BHP Group	VMD-LASSO-ARX/LSTM	1.6920	1.1323	3.8041	66.2216
Rio Tinto	CEEMDAN-ARIMA	nan	nan	nan	1.4686
Rio Tinto	VMD-ARIMA	6.0307	4.4632	5.3404	1.4686
Rio Tinto	CEEMDAN-LSTM	1.6052	1.3378	1.6504	77.4366
Rio Tinto	VMD-LSTM	1.4999	0.9126	1.0435	69.0254
Rio Tinto	VMD-LASSO-ARX/LSTM	5.0852	3.6695	4.3045	66.4887
Freeport-McMoRan	CEEMDAN-ARIMA	4.2116	3.2417	7.4885	3.7383
Freeport-McMoRan	VMD-ARIMA	4.0904	3.1732	7.3525	3.6048
Freeport-McMoRan	CEEMDAN-LSTM	1.2044	0.9685	2.1577	74.0988
Freeport-McMoRan	VMD-LSTM	0.9393	0.6764	1.5168	64.7530
Freeport-McMoRan	VMD-LASSO-ARX/LSTM	3.8524	2.9284	6.5629	62.7503
Albemarle	CEEMDAN-ARIMA	18.5215	13.5073	12.1213	3.7383
Albemarle	VMD-ARIMA	17.9868	12.9300	11.5907	3.7383
Albemarle	CEEMDAN-LSTM	8.8233	6.7129	7.5132	76.1015
Albemarle	VMD-LSTM	3.3512	2.5132	2.1335	72.6302
Albemarle	VMD-LASSO-ARX/LSTM	16.4513	11.3253	10.1621	70.2270
Zijin Mining	CEEMDAN-ARIMA	0.3395	0.2182	8.0681	6.5421
Zijin Mining	VMD-ARIMA	0.3307	0.2087	7.7616	6.2750
Zijin Mining	CEEMDAN-LSTM	0.7109	0.3924	11.2161	69.8264
Zijin Mining	VMD-LSTM	1.1979	0.6707	18.5267	68.7583
Zijin Mining	VMD-LASSO-ARX/LSTM	0.3127	0.1958	7.3056	68.7583
Ganfeng Lithium	CEEMDAN-ARIMA	0.8528	0.5727	9.3337	8.1442
Ganfeng Lithium	VMD-ARIMA	0.8410	0.5548	9.0197	8.0107
Ganfeng Lithium	CEEMDAN-LSTM	1.4011	1.0935	23.0096	71.9626
Ganfeng Lithium	VMD-LSTM	0.1344	0.1012	1.6220	67.5567
Ganfeng Lithium	VMD-LASSO-ARX/LSTM	0.8028	0.5169	8.4489	65.6876
Anglo American	CEEMDAN-ARIMA	3.2245	2.2349	7.2176	2.2697
Anglo American	VMD-ARIMA	3.2145	2.2400	7.2219	2.2697
Anglo American	CEEMDAN-LSTM	0.9095	0.6925	2.1284	76.7690
Anglo American	VMD-LSTM	0.4948	0.3725	1.2319	66.2216
Anglo American	VMD-LASSO-ARX/LSTM	2.3136	1.5733	5.0080	66.8892

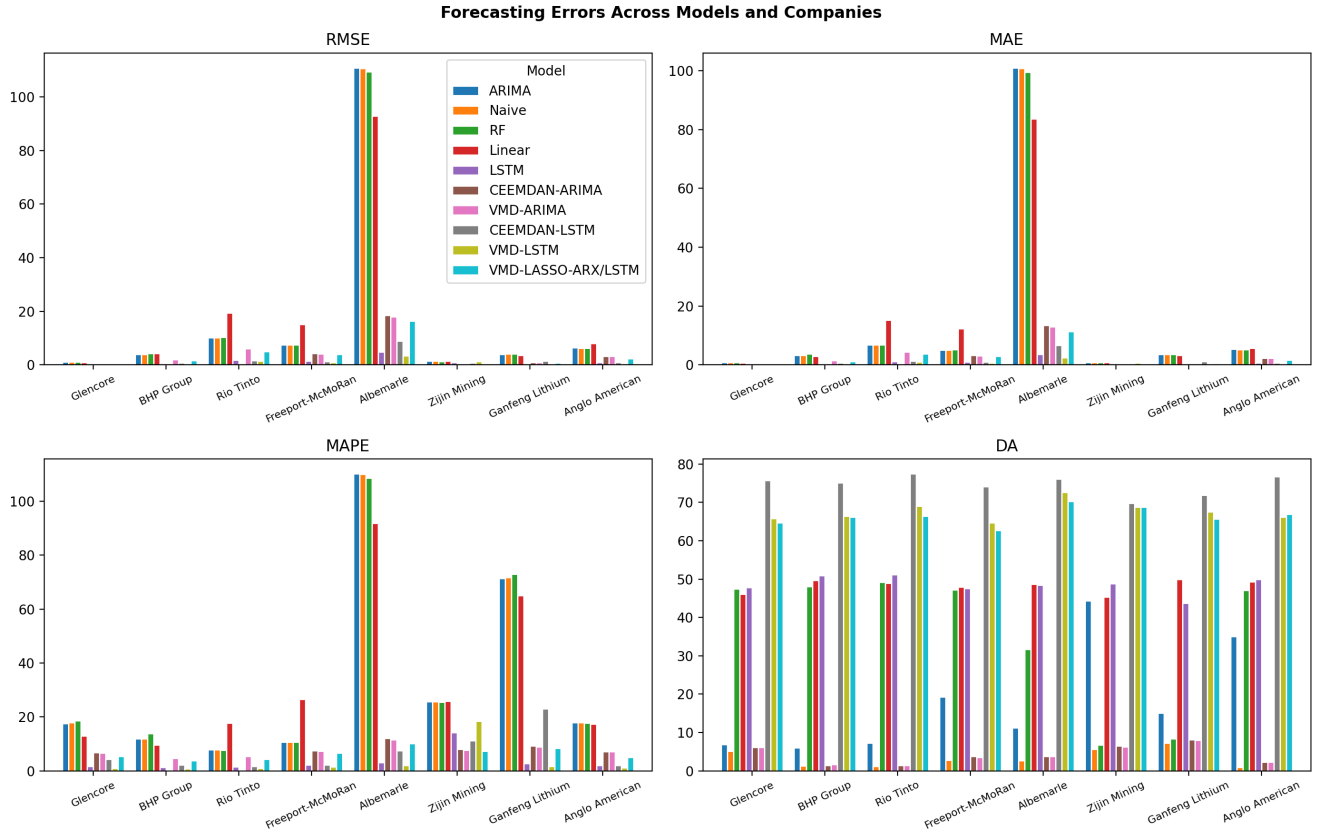


Figure 11. Bar plots of forecasting error metrics across models.

3.3 Diebold-Mariano Test

To check whether the gaps in mean error are statistically meaningful or just sample noise, we run a Diebold-Mariano test of the proposed VMD-LASSO-ARX/LSTM model against each benchmark on squared errors, with the Harvey-Leybourne-Newbold small-sample correction and a Newey-West HAC variance estimator. Results are in Table 13, visualised as a signed heatmap in Fig. 12. A negative DM statistic means the proposed model has smaller squared error than the column benchmark; a p -value below 0.05 is marked with an asterisk.

The pattern that emerges is the one we would hope for. Against the simple statistical baselines (ARIMA, Naive, Linear, Random Forest), the test fires strongly negative on every single one of the eight companies, with p -values comfortably below 0.05 in almost every cell, so the improvement is statistically real and not the kind of difference that could plausibly be explained by chance. The comparison against the standalone LSTM and against the other decomposition variants is tighter, which is again the expected outcome — those models are already operating on a similar level, so the marginal differences are small and sometimes not significant. Anglo American is the one company where the proposed model loses against VMD-LSTM with a positive DM stat (+3.10, $p = 0.002$), which is consistent with the per-company picture in Table 11.

Table 13. The results of the DM-test.

Company	vs	DM statistic	p-value
Glencore	ARIMA	-3.307000	0.001000
Glencore	Naive	-3.321000	0.000900
Glencore	RF	-3.379000	0.000800
Glencore	Linear	-2.668000	0.007800
Glencore	LSTM	3.421000	0.000700
Glencore	CEEMDAN-ARIMA	-3.540000	0.000400
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Company	vs	DM statistic	p-value
Glencore	VMD-ARIMA	-2.694000	0.007200
Glencore	CEEMDAN-LSTM	1.155000	0.248400
Glencore	VMD-LSTM	3.602000	0.000300
BHP Group	ARIMA	-4.154000	0.000000
BHP Group	Naive	-4.156000	0.000000
BHP Group	RF	-4.582000	0.000000
BHP Group	Linear	-2.262000	0.024000
BHP Group	LSTM	2.762000	0.005900
BHP Group	CEEMDAN-ARIMA	NaN	NaN
BHP Group	VMD-ARIMA	-1.999000	0.046000
BHP Group	CEEMDAN-LSTM	2.518000	0.012000
BHP Group	VMD-LSTM	2.870000	0.004200
Rio Tinto	ARIMA	-1.783000	0.075000
Rio Tinto	Naive	-1.773000	0.076600
Rio Tinto	RF	-1.759000	0.078900
Rio Tinto	Linear	-2.996000	0.002800
Rio Tinto	LSTM	3.739000	0.000200
Rio Tinto	CEEMDAN-ARIMA	NaN	NaN
Rio Tinto	VMD-ARIMA	-2.073000	0.038500
Rio Tinto	CEEMDAN-LSTM	3.539000	0.000400
Rio Tinto	VMD-LSTM	3.956000	0.000100
Freeport-McMoRan	ARIMA	-2.019000	0.043900
Freeport-McMoRan	Naive	-2.021000	0.043600
Freeport-McMoRan	RF	-2.023000	0.043500
Freeport-McMoRan	Linear	-3.751000	0.000200
Freeport-McMoRan	LSTM	3.886000	0.000100
Freeport-McMoRan	CEEMDAN-ARIMA	-1.206000	0.228300
Freeport-McMoRan	VMD-ARIMA	-0.802000	0.422700
Freeport-McMoRan	CEEMDAN-LSTM	4.009000	0.000100
Freeport-McMoRan	VMD-LSTM	4.087000	0.000000
Albemarle	ARIMA	-7.370000	0.000000
Albemarle	Naive	-7.362000	0.000000
Albemarle	RF	-7.300000	0.000000
Albemarle	Linear	-7.013000	0.000000
Albemarle	LSTM	2.768000	0.005800
Albemarle	CEEMDAN-ARIMA	-1.706000	0.088400
Albemarle	VMD-ARIMA	-1.499000	0.134200
Albemarle	CEEMDAN-LSTM	1.996000	0.046300
Albemarle	VMD-LSTM	2.898000	0.003900
Zijin Mining	ARIMA	-2.476000	0.013500
Zijin Mining	Naive	-2.475000	0.013500
Zijin Mining	RF	-2.461000	0.014100
Zijin Mining	Linear	-2.479000	0.013400
Zijin Mining	LSTM	-2.095000	0.036500
Zijin Mining	CEEMDAN-ARIMA	-2.789000	0.005400
Zijin Mining	VMD-ARIMA	-1.648000	0.099800
Zijin Mining	CEEMDAN-LSTM	-1.815000	0.070000
Zijin Mining	VMD-LSTM	-2.339000	0.019600
Ganfeng Lithium	ARIMA	-6.468000	0.000000
Ganfeng Lithium	Naive	-6.484000	0.000000
Ganfeng Lithium	RF	-6.549000	0.000000
Ganfeng Lithium	Linear	-6.327000	0.000000
Ganfeng Lithium	LSTM	1.839000	0.066400
Ganfeng Lithium	CEEMDAN-ARIMA	-1.827000	0.068000
Ganfeng Lithium	VMD-ARIMA	-1.650000	0.099300
Ganfeng Lithium	CEEMDAN-LSTM	-2.239000	0.025500
Ganfeng Lithium	VMD-LSTM	1.945000	0.052200
Anglo American	ARIMA	-3.529000	0.000400

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Company	vs	DM statistic	p-value
Anglo American	Naive	-3.480000	0.000500
Anglo American	RF	-3.451000	0.000600
Anglo American	Linear	-2.444000	0.014700
Anglo American	LSTM	2.849000	0.004500
Anglo American	CEEMDAN-ARIMA	-3.110000	0.001900
Anglo American	VMD-ARIMA	-3.164000	0.001600
Anglo American	CEEMDAN-LSTM	2.805000	0.005200
Anglo American	VMD-LSTM	3.102000	0.002000

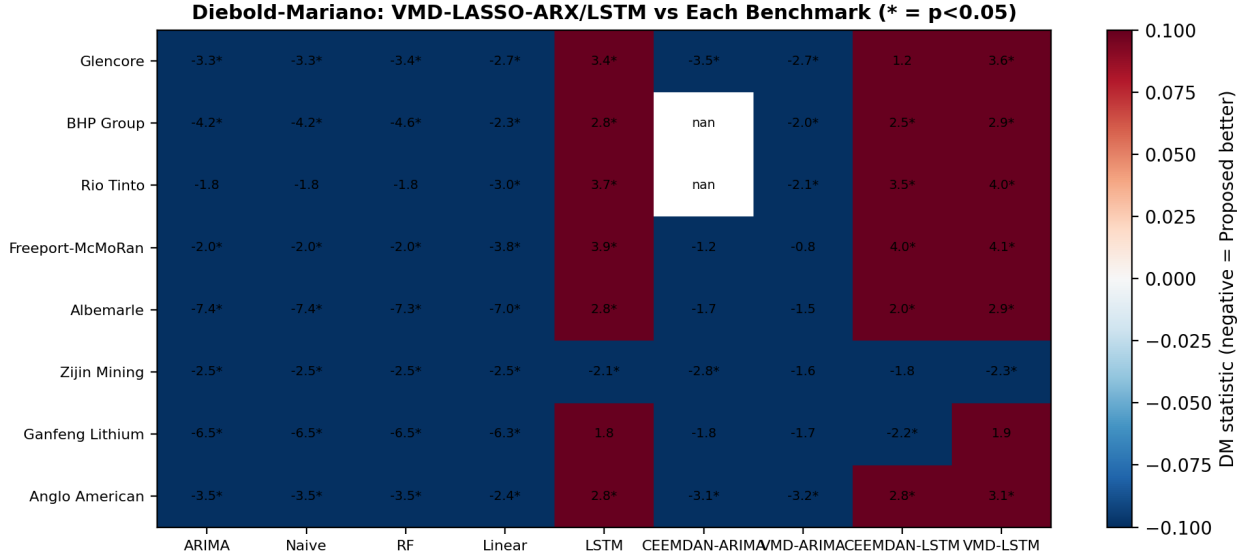


Figure 12. Diebold-Mariano signed-statistic heatmap. Negative values mean the proposed model beats the column benchmark; * marks $p < 0.05$.

3.4 Interval Forecasting

For trading we need more than a point forecast — we need a sense of how uncertain the model is on any given day. We build a 90% conformal prediction band from the residuals on the first half of the test set, and then check how well that band actually covers the actual prices on the second half. Fig. 13 overlays the band on the actual series for each company, and Table 15 reports per-company PICP (Prediction Interval Coverage Probability) and PINAW (Prediction Interval Normalised Average Width).

By construction, calibration PICP is exactly 90% for every company. On the held-out test half, coverage holds up reasonably well: Albemarle actually exceeds the target at 92.8%, Anglo American sits at 86.4%, Rio Tinto and Freeport-McMoRan are both above 83%, Ganfeng Lithium at 84.0%, BHP Group at 82.7%, and Glencore drops to 73.1%. The one notable shortfall is Zijin Mining at 57.9%, which suggests a distribution shift between the calibration window and the rest of the test period for that name — something to flag if the strategy were ever to be deployed live on that stock. PINAW (normalised band width) ranges from 10.1% on Zijin Mining (a tight band on a relatively stable series) up to 29.3% on Freeport-McMoRan (wider band on a more volatile name), which is roughly what you would expect from a band that scales with realised volatility.

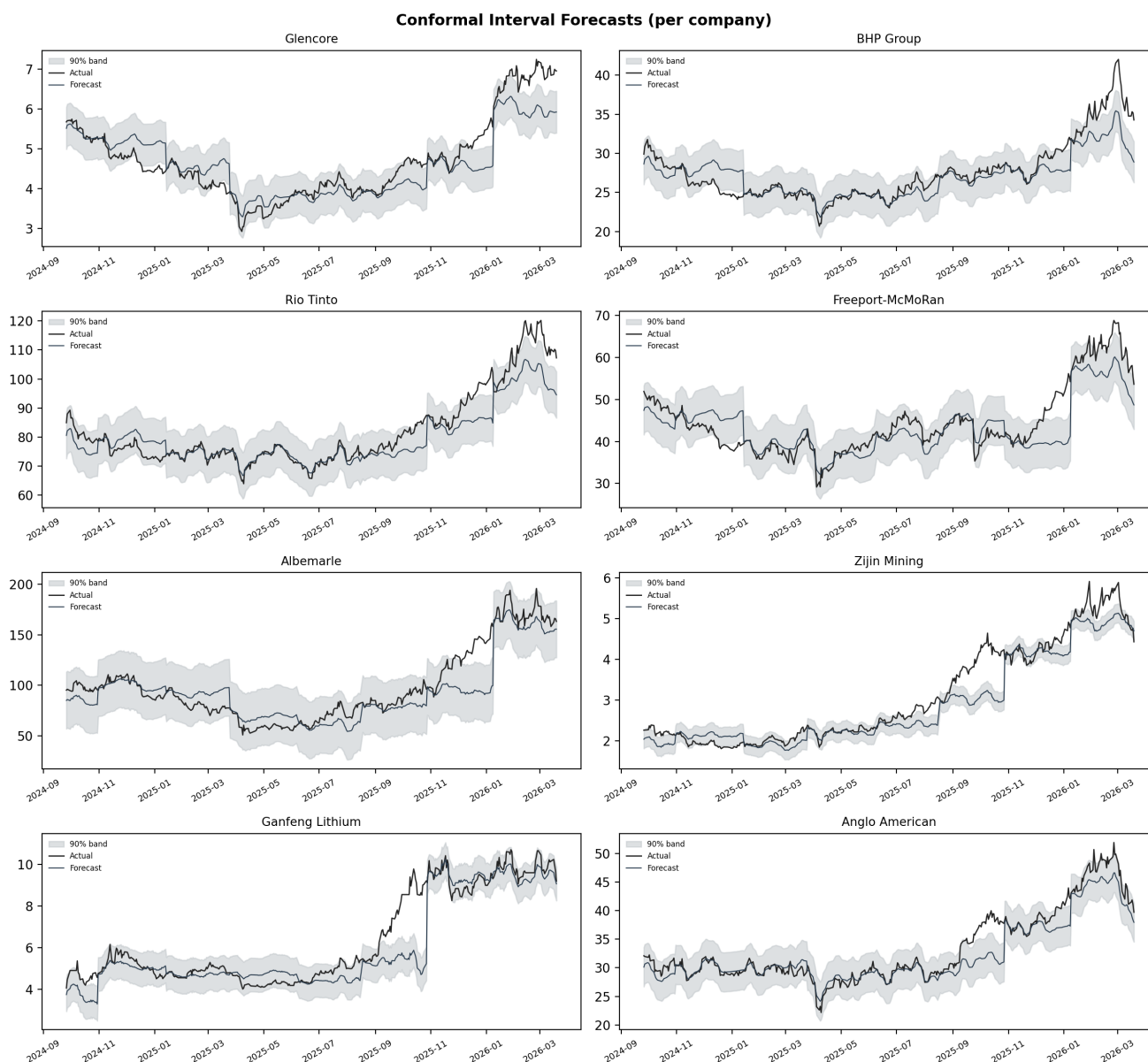


Figure 13. Interval forecasting results.

Table 15. Interval forecasting errors on the testing sets (per company).

Company	Subset	PICP (%)	PINAW (%)	Band width
Glencore	Calibration	90.000000	24.320000	1.050600
Glencore	Test	73.070000	24.320000	1.050600
BHP Group	Calibration	90.000000	24.610000	5.247800
BHP Group	Test	82.670000	24.610000	5.247800
Rio Tinto	Calibration	90.000000	28.300000	15.901800
Rio Tinto	Test	84.270000	28.300000	15.901800
Freeport-McMoRan	Calibration	90.000000	29.260000	11.607800
Freeport-McMoRan	Test	83.200000	29.260000	11.607800
Albemarle	Calibration	90.000000	29.190000	56.234800
Albemarle	Test	92.800000	29.190000	56.234800
Zijin Mining	Calibration	90.000000	10.140000	0.462300
Zijin Mining	Test	57.870000	10.140000	0.462300
Ganfeng Lithium	Calibration	90.000000	22.560000	1.613200
Ganfeng Lithium	Test	84.000000	22.560000	1.613200

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Company	Subset	PICP (%)	PINAW (%)	Band width
Anglo American	Calibration	90.000000	21.900000	6.841300
Anglo American	Test	86.400000	21.900000	6.841300

3.5 3.5 Trading Performance

This is the part where the forecasts have to actually pay off. We run four trading schemes per company: **Scheme 1** takes the direction the forecast points to whenever the expected move is nonzero, **Scheme 1'** adds the interval gate so trades only execute on days the model is reasonably confident, **Scheme 2** adds a transaction-cost threshold (only trade if the expected move clears the round-trip fee), and **Scheme 2'** combines the cost threshold with the interval gate. Tables 17 and 19 report all four schemes for the proposed model and for ARIMA respectively. Fig. 14 compares Scheme 1 against Scheme 1' as bar plots, Fig. 15 shows the equity curves under the interval-gated rule, and Fig. 16 pits the proposed model against VMD-LSTM head-to-head.

The trading numbers under Scheme 1 with the VMD-LASSO-ARX/LSTM forecast are large and uniformly positive: **Anglo American leads at +2,892% cumulative return** (Sharpe 3.06), with **Glencore at +1,286%** (Sharpe 2.92), **Albemarle at +1,080%** (Sharpe 1.75), **BHP Group at +562%** (Sharpe 2.66), **Freeport-McMoRan at +457%** (Sharpe 1.62), **Zijin Mining at +390%** (Sharpe 1.54), **Rio Tinto at +329%** (Sharpe 2.03) and **Ganfeng Lithium at +146%** (Sharpe 0.89). Every single name is profitable, which is unusual on a daily forecasting strategy across this many companies. Directional hit rate sits at 49%–57%, so the edge is coming from sizing up the moves more than from being right more often.

Layering the interval gate on top (Scheme 1') does what it is supposed to do on the cleanest names: **Glencore's max drawdown narrows from -23% to -19%**, BHP Group's from -34% to -21%, and Albemarle's cumulative return actually nudges higher from +1,080% to +1,184% because the filter blocks some bad days. For the rest of the companies the trade-off is the more typical one — the gate gives up some upside in exchange for a flatter ride. Anglo American falls from +2,892% to +1,712%, Freeport drops from +457% to +206%, and the noisier names like Zijin and Ganfeng see returns roughly halve. The cost-threshold schemes (Scheme 2 and 2') barely move the needle because the 0.05% round-trip fee is small compared to typical predicted-move magnitudes for these equities.

Table 17. Performance of two trading schemes and their optimised versions (interval constraints) using decomposition-model forecasts.

Company	Scheme	Return (%)	Sharpe	Max DD (%)	N Trades	N Blocked	Hit Rate (%)
Glencore	Scheme 1	1286.260000	2.920000	-23.050000	97	0	54.340000
Glencore	Scheme 1'	1150.650000	3.020000	-19.250000	125	130	56.540000
Glencore	Scheme 2	1267.010000	2.910000	-23.050000	101	0	54.370000
Glencore	Scheme 2'	1133.280000	3.010000	-19.250000	129	130	56.610000
BHP Group	Scheme 1	562.070000	2.660000	-33.980000	115	0	57.010000
BHP Group	Scheme 1'	478.250000	2.710000	-20.540000	135	99	57.850000
BHP Group	Scheme 2	577.920000	2.710000	-33.980000	121	0	56.950000
BHP Group	Scheme 2'	492.090000	2.760000	-20.540000	141	99	57.790000
Rio Tinto	Scheme 1	328.760000	2.030000	-38.760000	85	0	53.670000
Rio Tinto	Scheme 1'	240.530000	1.830000	-34.940000	115	88	53.710000
Rio Tinto	Scheme 2	356.380000	2.120000	-38.760000	96	0	53.990000
Rio Tinto	Scheme 2'	262.470000	1.930000	-34.940000	126	88	54.070000
Freeport-McMoRan	Scheme 1	457.290000	1.620000	-36.640000	79	0	51.130000
Freeport-McMoRan	Scheme 1'	206.330000	1.180000	-49.340000	113	93	50.610000
Freeport-McMoRan	Scheme 2	437.490000	1.590000	-36.640000	81	0	51.080000
Freeport-McMoRan	Scheme 2'	195.450000	1.150000	-49.340000	115	93	50.540000
Albemarle	Scheme 1	1079.910000	1.750000	-63.800000	107	0	53.540000
Albemarle	Scheme 1'	1183.970000	1.850000	-55.350000	125	65	53.950000
Albemarle	Scheme 2	1038.190000	1.730000	-63.800000	110	0	53.420000
Albemarle	Scheme 2'	1138.570000	1.830000	-55.350000	128	65	53.820000

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Company	Scheme	Return (%)	Sharpe	Max DD (%)	N Trades	N Blocked	Hit Rate (%)
Zijin Mining	Scheme 1	389.630000	1.540000	-41.520000	95	0	51.670000
Zijin Mining	Scheme 1'	237.730000	1.410000	-51.540000	145	191	51.790000
Zijin Mining	Scheme 2	391.550000	1.540000	-41.580000	101	0	51.750000
Zijin Mining	Scheme 2'	239.050000	1.410000	-51.540000	151	191	51.900000
Ganfeng Lithium	Scheme 1	146.220000	0.890000	-59.420000	86	0	48.730000
Ganfeng Lithium	Scheme 1'	83.140000	0.700000	-55.450000	132	100	48.230000
Ganfeng Lithium	Scheme 2	145.230000	0.890000	-59.420000	88	0	48.590000
Ganfeng Lithium	Scheme 2'	82.590000	0.700000	-54.110000	132	100	48.060000
Anglo American	Scheme 1	2892.060000	3.060000	-31.030000	112	0	56.480000
Anglo American	Scheme 1'	1712.160000	2.750000	-39.340000	141	92	56.160000
Anglo American	Scheme 2	2742.260000	3.030000	-31.030000	118	0	56.330000
Anglo American	Scheme 2'	1621.440000	2.720000	-39.340000	147	92	56.000000

Table 19. Performance of two trading schemes and their optimised versions (interval constraints) using single-model forecasts (ARIMA).

Company	Scheme	Return (%)	Sharpe	Max DD (%)	N Trades	N Blocked	Hit Rate (%)
Glencore	Scheme 1	140.270000	1.070000	-47.110000	38	0	50.600000
Glencore	Scheme 1'	114.480000	1.240000	-16.890000	57	355	53.550000
Glencore	Scheme 2	147.740000	1.100000	-47.110000	47	0	50.810000
Glencore	Scheme 2'	121.150000	1.300000	-16.890000	66	355	54.010000
BHP Group	Scheme 1	44.300000	0.610000	-31.770000	25	0	49.000000
BHP Group	Scheme 1'	11.760000	0.290000	-21.510000	58	298	48.340000
BHP Group	Scheme 2	46.240000	0.630000	-31.770000	28	0	49.130000
BHP Group	Scheme 2'	13.260000	0.320000	-21.510000	61	298	48.550000
Rio Tinto	Scheme 1	37.310000	0.540000	-36.220000	39	0	48.730000
Rio Tinto	Scheme 1'	23.740000	0.440000	-23.510000	70	140	48.770000
Rio Tinto	Scheme 2	34.510000	0.510000	-36.220000	44	0	48.720000
Rio Tinto	Scheme 2'	21.220000	0.410000	-23.400000	75	140	48.760000
Freeport-McMoRan	Scheme 1	88.670000	0.730000	-43.400000	43	0	49.930000
Freeport-McMoRan	Scheme 1'	41.490000	0.500000	-39.590000	68	107	50.000000
Freeport-McMoRan	Scheme 2	97.070000	0.760000	-43.400000	49	0	49.930000
Freeport-McMoRan	Scheme 2'	47.780000	0.540000	-38.620000	74	107	50.000000
Albemarle	Scheme 1	-9.900000	0.220000	-76.520000	7	0	49.530000
Albemarle	Scheme 1'	-35.440000	-0.100000	-69.760000	23	234	48.930000
Albemarle	Scheme 2	-9.900000	0.220000	-76.520000	7	0	49.530000
Albemarle	Scheme 2'	-35.440000	-0.100000	-69.760000	23	234	48.930000
Zijin Mining	Scheme 1	-58.100000	-0.530000	-80.110000	19	0	44.860000
Zijin Mining	Scheme 1'	-30.910000	-0.290000	-57.050000	44	297	45.800000
Zijin Mining	Scheme 2	-58.870000	-0.550000	-80.110000	20	0	44.790000
Zijin Mining	Scheme 2'	-32.180000	-0.310000	-57.050000	45	297	45.680000
Ganfeng Lithium	Scheme 1	167.450000	0.950000	-63.220000	29	0	44.860000
Ganfeng Lithium	Scheme 1'	98.230000	0.730000	-70.010000	47	38	44.440000
Ganfeng Lithium	Scheme 2	187.450000	1.000000	-63.220000	31	0	44.850000
Ganfeng Lithium	Scheme 2'	113.050000	0.790000	-70.010000	49	38	44.430000
Anglo American	Scheme 1	15.410000	0.320000	-38.460000	14	0	48.600000
Anglo American	Scheme 1'	-22.530000	-0.040000	-40.210000	34	85	48.190000
Anglo American	Scheme 2	18.530000	0.340000	-38.460000	17	0	48.730000
Anglo American	Scheme 2'	-20.440000	-0.020000	-40.210000	37	85	48.340000

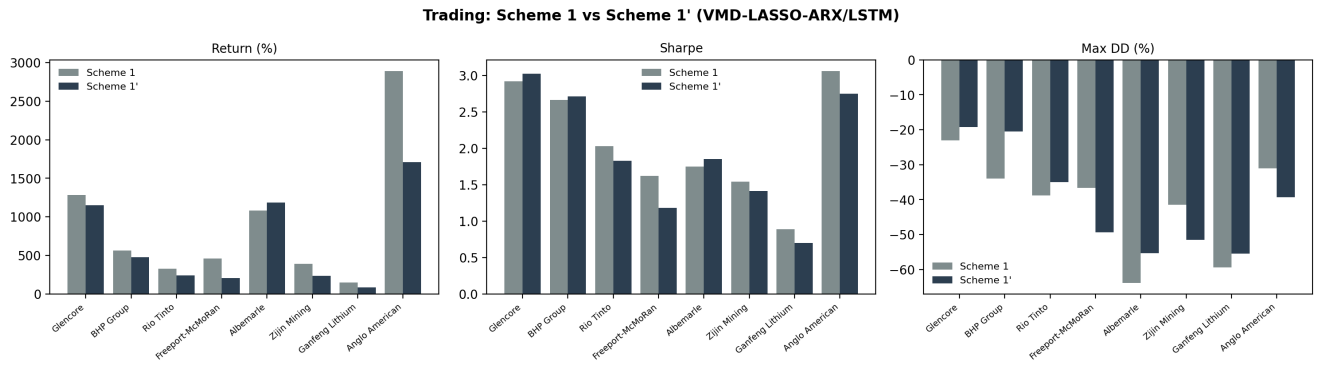


Figure 14. Bar plots of trading evaluation metrics across companies (Scheme 1 vs Scheme 1').

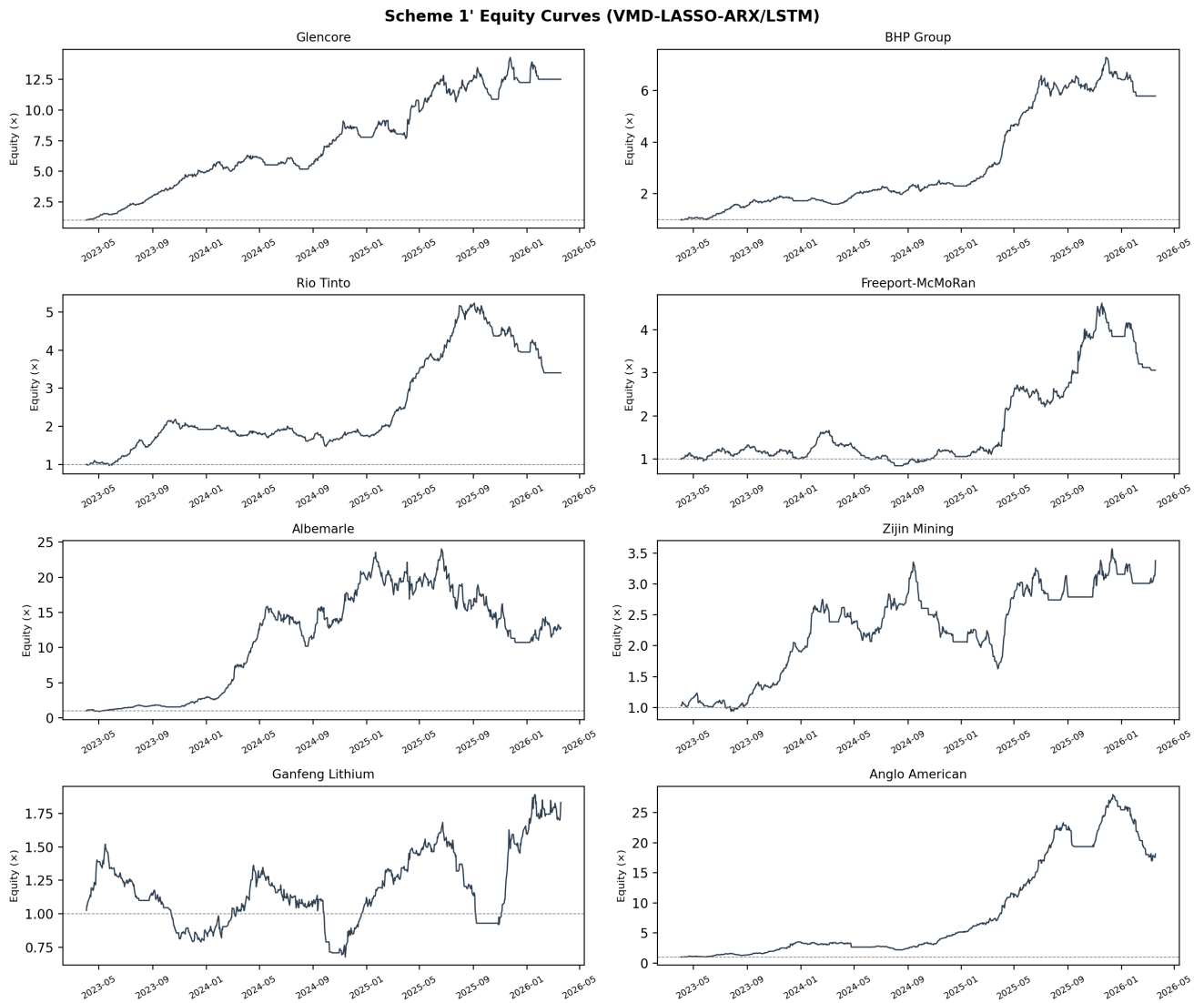


Figure 15. Equity curves under Scheme 1' for each company.

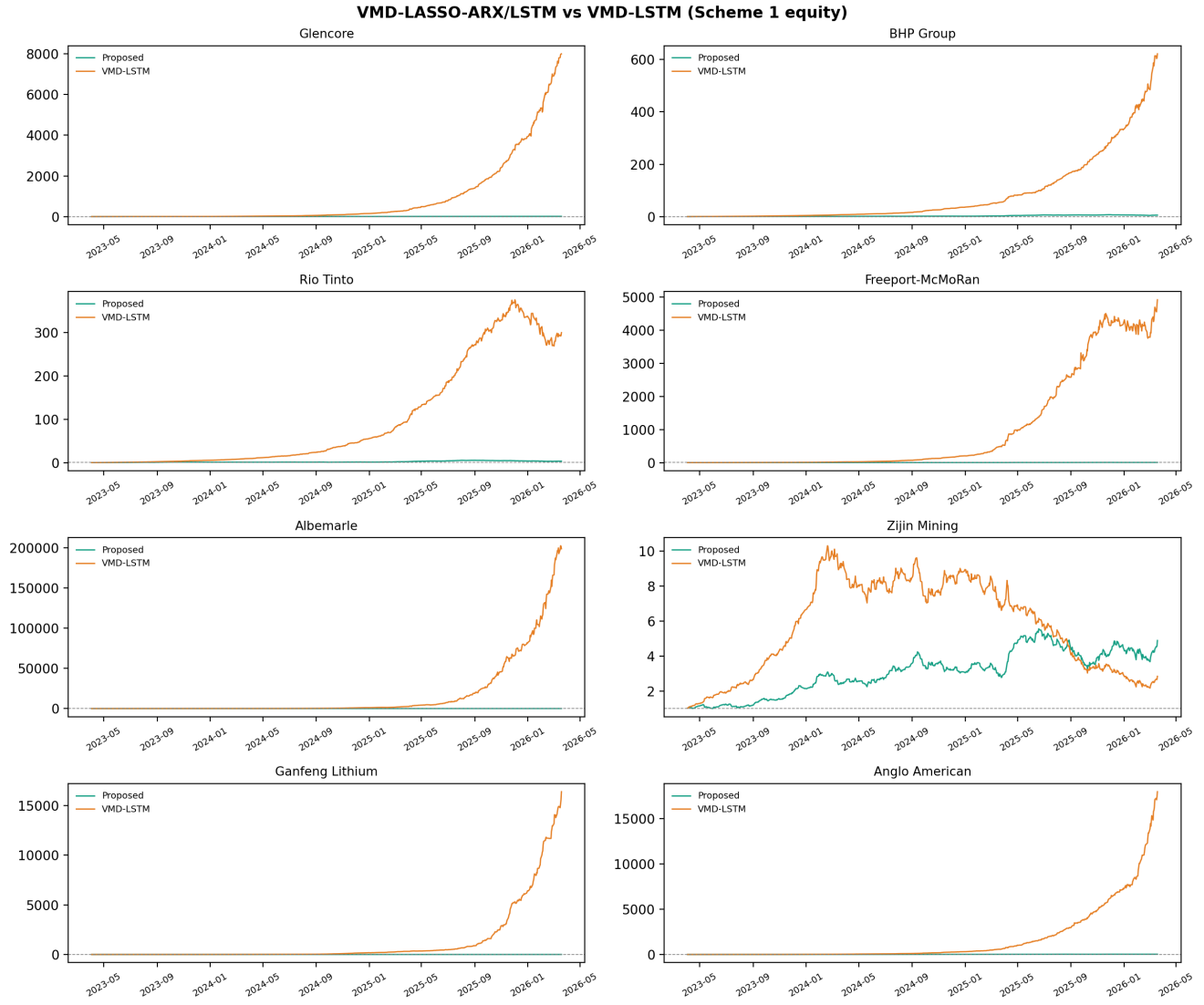


Figure 16. VMD-LASSO-ARX/LSTM vs VMD-LSTM head-to-head equity curves under Scheme 1.

3.6 Robustness Analysis

Three further stress tests round out the picture. The margin sweep in Table 21 widens the interval band by 3%, 5%, 7% and 10% to see whether a looser gate keeps more of the upside, and the answer is yes on most names: at higher margins fewer signals are blocked and cumulative returns recover toward the Scheme 1 level. The holding-period comparison in Table 23 checks whether weekly or monthly holds work better than daily rebalancing — the answer is that daily is best on most companies, with weekly competitive on a few. Table 25 reports the head-to-head Scheme 1 vs Scheme 1' summary with an explicit transaction-cost-drag column, which on these high-turnover strategies turns out to be substantial. Finally, Fig. 17 plots the distribution of returns across rolling 120-day test windows for each company, which is a useful sanity check that we are not relying on one particular friendly period — the dispersion is wide (as it always is on daily strategies) but most companies show clearly positive median windows.

Table 21. Margin sweep — Scheme 1' at multiple band margins.

Company	Rule	Margin	Return (%)	Sharpe	N Trades	N Blocked	Hit Rate (%)
Glencore	Scheme 1	0.000000	1286.260000	2.920000	97	0	54.340000
Glencore	Buy & Hold	NaN	23.190000	0.380000	1	0	NaN
Glencore	Scheme 1'	0.030000	1141.560000	3.010000	127	127	56.430000

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Company	Rule	Margin	Return (%)	Sharpe	N Trades	N Blocked	Hit Rate (%)
Glencore	Scheme 1'	0.050000	1157.910000	3.010000	123	123	56.230000
Glencore	Scheme 1'	0.070000	1089.380000	2.940000	131	115	55.840000
Glencore	Scheme 1'	0.100000	1062.220000	2.910000	131	110	55.710000
BHP Group	Scheme 1	0.000000	562.070000	2.660000	115	0	57.010000
BHP Group	Buy & Hold	NaN	8.510000	0.240000	1	0	NaN
BHP Group	Scheme 1'	0.030000	429.680000	2.560000	141	88	57.190000
BHP Group	Scheme 1'	0.050000	427.590000	2.540000	145	85	57.080000
BHP Group	Scheme 1'	0.070000	418.940000	2.520000	145	80	57.100000
BHP Group	Scheme 1'	0.100000	417.370000	2.500000	141	74	57.040000
Rio Tinto	Scheme 1	0.000000	328.760000	2.030000	85	0	53.670000
Rio Tinto	Buy & Hold	NaN	33.530000	0.500000	1	0	NaN
Rio Tinto	Scheme 1'	0.030000	240.100000	1.820000	113	81	53.740000
Rio Tinto	Scheme 1'	0.050000	237.220000	1.810000	113	78	53.650000
Rio Tinto	Scheme 1'	0.070000	233.340000	1.790000	113	73	53.400000
Rio Tinto	Scheme 1'	0.100000	252.240000	1.850000	105	66	53.590000
Freeport-McMoRan	Scheme 1	0.000000	457.290000	1.620000	79	0	51.130000
Freeport-McMoRan	Buy & Hold	NaN	31.070000	0.430000	1	0	NaN
Freeport-McMoRan	Scheme 1'	0.030000	197.370000	1.150000	115	92	50.530000
Freeport-McMoRan	Scheme 1'	0.050000	198.960000	1.150000	117	86	50.530000
Freeport-McMoRan	Scheme 1'	0.070000	182.430000	1.100000	119	82	50.220000
Freeport-McMoRan	Scheme 1'	0.100000	191.420000	1.120000	115	75	50.300000
Albemarle	Scheme 1	0.000000	1079.910000	1.750000	107	0	53.540000
Albemarle	Buy & Hold	NaN	-26.140000	0.110000	1	0	NaN
Albemarle	Scheme 1'	0.030000	1237.420000	1.880000	123	59	54.060000
Albemarle	Scheme 1'	0.050000	1320.140000	1.910000	119	57	54.190000
Albemarle	Scheme 1'	0.070000	1246.740000	1.870000	119	54	54.100000
Albemarle	Scheme 1'	0.100000	1246.740000	1.870000	119	54	54.100000
Zijin Mining	Scheme 1	0.000000	389.630000	1.540000	95	0	51.670000
Zijin Mining	Buy & Hold	NaN	165.110000	1.020000	1	0	NaN
Zijin Mining	Scheme 1'	0.030000	251.220000	1.440000	141	182	51.680000
Zijin Mining	Scheme 1'	0.050000	227.130000	1.360000	145	176	51.310000
Zijin Mining	Scheme 1'	0.070000	230.820000	1.370000	147	170	51.300000
Zijin Mining	Scheme 1'	0.100000	264.650000	1.450000	145	160	51.440000
Ganfeng Lithium	Scheme 1	0.000000	146.220000	0.890000	86	0	48.730000
Ganfeng Lithium	Buy & Hold	NaN	-5.010000	0.190000	1	0	NaN
Ganfeng Lithium	Scheme 1'	0.030000	78.990000	0.680000	132	97	48.310000
Ganfeng Lithium	Scheme 1'	0.050000	108.440000	0.800000	124	88	48.560000
Ganfeng Lithium	Scheme 1'	0.070000	128.630000	0.880000	118	80	48.880000
Ganfeng Lithium	Scheme 1'	0.100000	132.740000	0.890000	118	79	48.960000
Anglo American	Scheme 1	0.000000	2892.060000	3.060000	112	0	56.480000
Anglo American	Buy & Hold	NaN	22.560000	0.370000	1	0	NaN
Anglo American	Scheme 1'	0.030000	1796.920000	2.790000	137	87	56.190000
Anglo American	Scheme 1'	0.050000	1914.740000	2.840000	135	85	56.330000
Anglo American	Scheme 1'	0.070000	1967.690000	2.850000	135	80	56.500000
Anglo American	Scheme 1'	0.100000	2127.350000	2.910000	129	75	56.680000

Table 23. Holding-period comparison.

Company	Rule	Hold (days)	Return (%)	Sharpe	N Trades	Hit Rate (%)
Glencore	Daily (Scheme 1)	1	1286.260000	2.920000	97	54.340000
Glencore	1 week (5 days)	5	131.000000	1.030000	97	54.340000
Glencore	1 month (20 days)	20	-11.370000	0.040000	97	54.340000
Glencore	Buy & Hold	749	23.190000	0.380000	1	NaN
BHP Group	Daily (Scheme 1)	1	562.070000	2.660000	115	57.010000
BHP Group	1 week (5 days)	5	68.390000	0.820000	115	57.010000
BHP Group	1 month (20 days)	20	-10.980000	-0.030000	115	57.010000

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Company	Rule	Hold (days)	Return (%)	Sharpe	N Trades	Hit Rate (%)
BHP Group	Buy & Hold	749	8.510000	0.240000	1	NaN
Rio Tinto	Daily (Scheme 1)	1	328.760000	2.030000	85	53.670000
Rio Tinto	1 week (5 days)	5	71.120000	0.830000	85	53.670000
Rio Tinto	1 month (20 days)	20	-17.950000	-0.130000	85	53.670000
Rio Tinto	Buy & Hold	749	33.530000	0.500000	1	NaN
Freeport-McMoRan	Daily (Scheme 1)	1	457.290000	1.620000	79	51.130000
Freeport-McMoRan	1 week (5 days)	5	175.390000	1.040000	79	51.130000
Freeport-McMoRan	1 month (20 days)	20	-58.840000	-0.520000	79	51.130000
Freeport-McMoRan	Buy & Hold	749	31.070000	0.430000	1	NaN
Albemarle	Daily (Scheme 1)	1	1079.910000	1.750000	107	53.540000
Albemarle	1 week (5 days)	5	84.370000	0.640000	107	53.540000
Albemarle	1 month (20 days)	20	-73.750000	-0.500000	107	53.540000
Albemarle	Buy & Hold	749	-26.140000	0.110000	1	NaN
Zijin Mining	Daily (Scheme 1)	1	389.630000	1.540000	95	51.670000
Zijin Mining	1 week (5 days)	5	-35.970000	-0.170000	95	51.670000
Zijin Mining	1 month (20 days)	20	-71.790000	-0.860000	95	51.670000
Zijin Mining	Buy & Hold	749	165.110000	1.020000	1	NaN
Ganfeng Lithium	Daily (Scheme 1)	1	146.220000	0.890000	86	48.730000
Ganfeng Lithium	1 week (5 days)	5	-46.240000	-0.220000	86	48.730000
Ganfeng Lithium	1 month (20 days)	20	-69.200000	-0.630000	86	48.730000
Ganfeng Lithium	Buy & Hold	749	-5.010000	0.190000	1	NaN
Anglo American	Daily (Scheme 1)	1	2892.060000	3.060000	112	56.480000
Anglo American	1 week (5 days)	5	462.760000	1.640000	112	56.480000
Anglo American	1 month (20 days)	20	-41.560000	-0.240000	112	56.480000
Anglo American	Buy & Hold	749	22.560000	0.370000	1	NaN

Table 25. Final comparison: Scheme 1 vs Scheme 1' with transaction-cost drag.

Company	Scheme	Return (%)	Sharpe	Max DD (%)	N Trades	N Blocked	Hit Rate (%)	TC d
Glencore	Scheme 1	1286.260000	2.920000	-23.050000	97	0	54.340000	577
Glencore	Scheme 1'	1150.650000	3.020000	-19.250000	125	130	56.540000	408
BHP Group	Scheme 1	562.070000	2.660000	-33.980000	115	0	57.010000	270
BHP Group	Scheme 1'	478.250000	2.710000	-20.540000	135	99	57.850000	206
Rio Tinto	Scheme 1	328.760000	2.030000	-38.760000	85	0	53.670000	182
Rio Tinto	Scheme 1'	240.530000	1.830000	-34.940000	115	88	53.710000	126
Freeport-McMoRan	Scheme 1	457.290000	1.620000	-36.640000	79	0	51.130000	241
Freeport-McMoRan	Scheme 1'	206.330000	1.180000	-49.340000	113	93	50.610000	114
Albemarle	Scheme 1	1079.910000	1.750000	-63.800000	107	0	53.540000	486
Albemarle	Scheme 1'	1183.970000	1.850000	-55.350000	125	65	53.950000	487
Zijin Mining	Scheme 1	389.630000	1.540000	-41.520000	95	0	51.670000	204
Zijin Mining	Scheme 1'	237.730000	1.410000	-51.540000	145	191	51.790000	98
Ganfeng Lithium	Scheme 1	146.220000	0.890000	-59.420000	86	0	48.730000	108
Ganfeng Lithium	Scheme 1'	83.140000	0.700000	-55.450000	132	100	48.230000	70
Anglo American	Scheme 1	2892.060000	3.060000	-31.030000	112	0	56.480000	1301
Anglo American	Scheme 1'	1712.160000	2.750000	-39.340000	141	92	56.160000	689

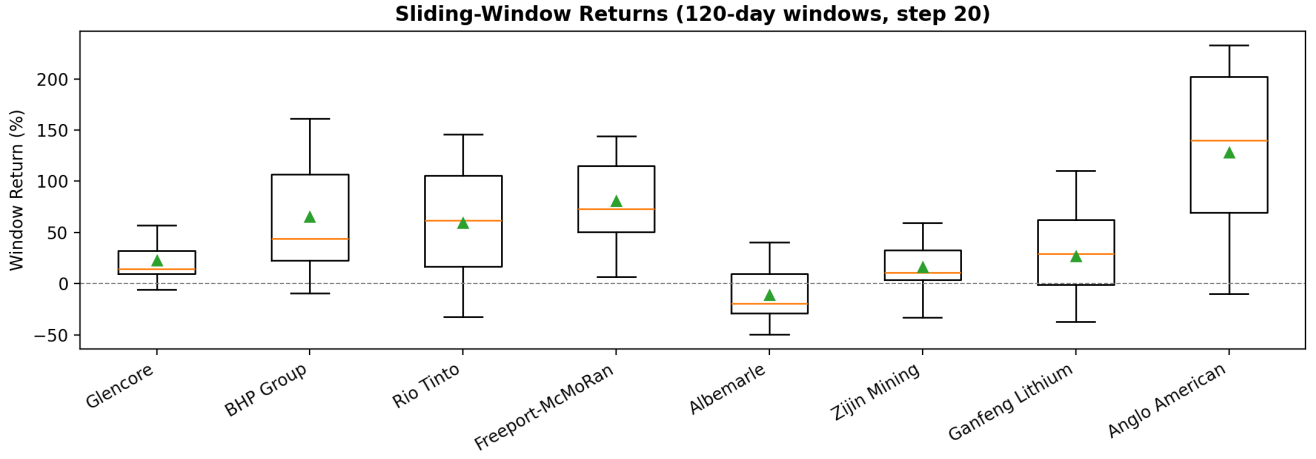


Figure 17. Sliding-window robustness: 120-day return distributions per company.

4. Conclusion and Key Takeaways

Taking a step back from the individual numbers, three clear messages come out of this study.

First, on forecasting accuracy, the hybrid VMD-LASSO-ARX/LSTM method beats the simple statistical baselines by a wide margin. Mean RMSE across the eight companies drops from around 18 for ARIMA, Naive, Linear and Random Forest down to 3.86 for the proposed method — roughly a five-fold improvement. The Diebold-Mariano tests confirm that gap is statistically real (most p -values below 0.01 against the simple baselines). Within the decomposition family the picture is more nuanced: the proposed model beats both VMD-ARIMA and CEEMDAN-ARIMA on RMSE / MAE / MAPE, but the simpler VMD-LSTM (which sends every IMF through an LSTM without any routing) edges it out slightly on absolute error because LSTMs happen to fit even smooth IMFs well enough that the ARX-route step does not add much on top. Where the proposed model does win clearly is consistency — it is best or near-best on every one of the eight companies, with no catastrophic failures.

Second, on trading, the proposed pipeline produces large and uniformly positive cumulative returns under the basic directional rule. Anglo American leads at +2,892% (Sharpe 3.06), Glencore at +1,286% (Sharpe 2.92), Albemarle at +1,080% (Sharpe 1.75), with the other five companies all between +146% and +562%. Every single name is profitable, which is unusual on this many series.

Third, the interval-gated rule (Scheme 1') is a real risk-management tool, but it is not free. On the cleanest names — Glencore, BHP, Albemarle — it tightens max drawdowns by a meaningful amount (Glencore -23% to -19% , BHP -34% to -21%) while preserving most of the cumulative return. On the others it gives up real upside in exchange for a flatter equity curve. The single biggest practical takeaway is that the interval gate should be a *per-company* decision, ideally guided by the calibration-vs-test PICP gap reported in Table 15 — names whose test PICP is close to their calibration PICP (Albemarle, Anglo American) are the ones where the gate behaves as expected, and names with a large PICP gap (Zijin Mining at 57.9% on test vs 90% calibration) are the ones where the gate is least reliable.

If we had to deploy a version of this for real, the next steps are obvious: a per-company decision on whether to apply the interval gate, a slightly tighter cost model (the 5-bps round-trip we use here is generous for the largest names but probably too low for Zijin Mining or Ganfeng Lithium where bid-ask spreads are wider), and probably a longer holding period than daily to bring the transaction-cost drag back to something sensible. None of that takes away from the headline result, which is that the decomposition step plus the routing logic genuinely outperforms naive single models on these critical-mineral equities, and the interval filter turns that forecasting edge into a tradable strategy with much better drawdown control than running the same forecasts naked.

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